

RESEARCH INTERESTS

My research focuses the foundation of trustworthy machine learning for safety-critical real-world AI, including agentic and physical AI by developing principled learning algorithms for trustworthy adaptation in the real world.

EDUCATION

University of Pennsylvania Philadelphia, USA
Ph.D. in Computer and Information Science 2021

- Advisors: Insup Lee and Osbert Bastani
- Thesis: *Uncertainty Estimation Toward Safe AI*
- Committee: Kostas Daniilidis, Nikolai Matni, Edgar Dobriban, and Kilian Q. Weinberger

Seoul National University Seoul, Korea
M.S. in Electrical and Computer Engineering 2012

- Advisor: Kyoung Mu Lee
- Thesis: *Abnormal Object Detection by Transformed-Canonical Scene Generation*

Seoul National University Seoul, Korea
B.S. in Computer Science and Engineering 2010

- Thesis Advisor: Byoung-Tak Zhang
- Thesis: *Behavioral Intelligence for Crowd Avatar in 3D Virtual Worlds*

EMPLOYMENT

POSTECH Pohang, Korea
Assistant Professor Aug. 2023-Now

Georgia Institute of Technology Atlanta, USA
Postdoctoral Researcher (Mentor: Taesoo Kim) Sept. 2021-July 2023

Google Cloud AI Sunnyvale, USA
Research Intern (Host: Kihyuk Sohn) Summer 2020

Biointelligence Laboratory, Seoul National University Seoul, Korea
Undergraduate Researcher 2008-2010

Republic of Korea Army Korea
Military Service 2006-2008

PUBLICATIONS

- [1] M. Lee*, Y. Jung*, and **S. Park**, “Online conformal abstention under adversarial bandit feedback”, in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, Sep. 2026.
- [2] J. Park, N. Park, S. Lee, Y.-C. Jhi, J. Cho, and **S. Park**, “Mj: Multi-turn llm jailbreaking via decomposed credit assignment”, in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, Sep. 2026.
- [3] M. Park*, S. Lee*, K. Kim, J. Heo, S. Cho, S. Li, **S. Park**, and D. Kim, “Transductive generalization for GNNs via optimal transport”, in *The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS)*, Sep. 2026.
- [4] J. Choi*, J. Yang*, and **S. Park**, “SAUL: Sharpness-aware augmented-lagrangian unlearning”, in *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)*, Aug. 2026.
- [5] K. Kim*, Y. Choi*, S. Lee, S. Jun, D. Kim†, and S. Park†, “The interplay of harness design and post-training in llm agents”, in *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)*, ICML26’ RLxF Workshop, Aug. 2026.
- [6] S. Lee, H. Kim, and **S. Park**, “Can we stop malicious ai? killbench: A benchmark for external ai kill switch feasibility”, in *The 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP Findings)*, Aug. 2026.
- [7] X. Cheng, F. Sang, Y. Yang, H. Zhang, **S. Park**, X. Zhang, and T. Kim, “The Illusion of Rust Safety: Detecting Modular Unsafe Functions with LLMs”, in *The ACM Conference on Computer and Communications Security (CCS)*, May 2026.
- [8] J. Jeong, T. Kim, and **S. Park**, “Towards functional correctness of large code models with selective generation”, in *International Conference on Machine Learning (ICML)*, NeurIPS’25 DL4C Workshop, May 2026.
- [9] J. Hwang, **S. Park**, and J. Ok, “LLM watermark evasion via bias inversion”, in *International Conference on Machine Learning (ICML)*, May 2026.
- [10] X. Cheng*, S. Park*, H. Han, X. Zhang, and T. Kim, “Ruby: Unmasking Unsafe Rust in Stripped Binaries via Machine Learning”, in *International Conference on Dependable Systems and Networks (DSN)*, Apr. 2026.
- [11] J. Yang, K. Kim, and **S. Park**, “Online conformal prediction with adversarial feedback via regret minimization”, in *The Fourteenth International Conference on Learning Representations (ICLR)*, Jan. 2026.
- [12] J. Hwang, J. Park, H. Park, D. Kim, **S. Park**, and J. Ok, “Retrieval-augmented generation with estimation of source reliability”, in *Empirical Methods in Natural Language Processing (EMNLP)*, 2025.
- [13] K. Kim, Y. Choi, H. Kim, D. Kim, and **S. Park**, “Chronobias: A benchmark for evaluating time-conditional group bias in the time-sensitive knowledge of large language models”, in *Empirical Methods in Natural Language Processing (EMNLP Findings)*, 2025.
- [14] S. Moon, M. Lee, **S. Park**, and D. Kim, “Holistic unlearning benchmark: A multi-faceted evaluation for text-to-image diffusion model unlearning”, in *International Conference on Computer Vision (ICCV)*, Oct. 2025.
- [15] H. Wang, Z. Yang, **S. Park**, Y. Yang, S. Kim, W. Lunardi, M. Andreoni, T. Kim, and W. Lee, “SOUNDBOOST: Effective RCA and Attack Detection for UAV via Acoustic Side-Channel”, in *Proceedings of the 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2025.

- [16] M. Lee, K. Kim, T. Kim, and **S. Park**, “Selective Generation for Controllable Language Models”, in *Neural Information Processing Systems (NeurIPS)*, 2024.
- [17] S. Li, **S. Park**, I. Lee, and O. Bastani, “TRAQ: Trustworthy Retrieval Augmented Question Answering via Conformal Prediction”, in *Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.
- [18] H. Park, J. Hwang, S. Mun, **S. Park**, and J. Ok, “MedBN: Robust Test Time Adaptation against Malicious Test Samples”, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [19] W. Si, **S. Park**, I. Lee, E. Dobriban, and O. Bastani, “PAC prediction sets under label shift”, in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [20] W. Si, S. Li, **S. Park**, I. Lee, and O. Bastani, “Angelic Patches for Improving Third-Party Object Detector Performance”, in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [21] **S. Park**, O. Bastani, and T. Kim, “ACon²: Adaptive Conformal Consensus for Provable Blockchain Oracles”, in *Proceedings of the 32nd USENIX Security Symposium (Security)*, 2023.
- [22] R. Kaur, K. Sridhar, **S. Park**, Y. Yang, S. Jha, A. Roy, O. Sokolsky, and I. Lee, “CODiT: Conformal out-of-distribution detection in time-series data for cyber-physical systems”, in *Proceedings of the 14th ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, 2023.
- [23] **S. Park**, X. Cheng, and T. Kim, “Unsafe’s Betrayal: Abusing Unsafe Rust in Binary Reverse Engineering via Machine Learning”, *arXiv preprint arXiv:2211.00111*, 2023.
- [24] **S. Park**, E. Dobriban, I. Lee, and O. Bastani, “PAC Prediction Sets for Meta-Learning”, in *Neural Information Processing Systems (NeurIPS)*, 2022.
- [25] S. Li, **S. Park**, X. Ji, I. Lee, and O. Bastani, “Towards PAC multi-object detection and tracking”, *arXiv preprint arXiv:2204.07482*, 2022.
- [26] S. Jang, **S. Park**, I. Lee, and O. Bastani, “Sequential covariate shift detection using classifier two-sample tests”, in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- [27] R. Kaur, S. Jha, A. Roy, **S. Park**, E. Dobriban, O. Sokolsky, and I. Lee, “iDECODE: In-distribution equivariance for conformal out-of-distribution detection”, in *Association for the Advancement of Artificial Intelligence (AAAI)*, 2021.
- [28] **S. Park**, S. Li, I. Lee, and O. Bastani, “PAC confidence predictions for deep neural network classifiers”, in *International Conference on Learning Representations (ICLR)*, 2021.
- [29] **S. Park**, O. Bastani, J. Weimer, and I. Lee, “Calibrated prediction with covariate shift via unsupervised domain adaptation”, in *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2020.
- [30] **S. Park**, O. Bastani, N. Matni, and I. Lee, “PAC confidence sets for deep neural networks via calibrated prediction”, in *International Conference on Learning Representations (ICLR)*, 2020.
- [31] **S. Park**, R. Ivanov, J. Weimer, and I. Lee, “From verification to learning for defense against adversarial examples in neural networks”, *Korea Cyber-security Competition*, 2018.
- [32] **S. Park**, J. Weimer, and I. Lee, “Resilient linear classification: An approach to deal with attacks on training data”, in *International Conference on Cyber-Physical Systems (ICCPS)*, 2017.
- [33] J. Oh, T. M. Howard, M. R. Walter, D. Barber, M. Zhu, **S. Park**, A. Suppe, L. Navarro-Serment, F. Duvallet, A. Boularias, *et al.*, “Integrated intelligence for human-robot teams”, in *International Symposium on Experimental Robotics (ISER)*, 2016.
- [34] **S. Park**, W. Kim, and K. M. Lee, “Abnormal object detection by canonical scene-based contextual model”, in *European Conference on Computer Vision (ECCV)*, 2012.

SCHOLARSHIPS AND AWARDS

- Best Paper Award from CKAIA 2026
- DARPA AIxCC Winner as Team Atlanta (\$4M Team Prize) 2025
- Best Paper Finalist from CKAIA 2025
- Bang Seung-Yang Graduate Fellowship from POSTECH CSE (awardee: Jaewoo Jeong) 2025
- POSTECH GSAI BK21 Best Paper Award 2025
- NeurIPS'24 Spotlight Paper (top 2.08%) 2024
- DARPA AIxCC Finalists as Team Atlanta (\$2M Team Prize) 2024
- NeurIPS'23 Outstanding Reviewer Award 2023
- ICML'23 TEACH Workshop Best Paper Award 2023
- ICCPS'23 Best Paper Award finalist 2023
- NeurIPS'21 Outstanding Reviewer Award (top 8% of reviewers) 2021
- Korea cyber-security paper competition Best Paper Award (\$4,500) 2018
- PhD fellowship at University of Pennsylvania 2013-2021
- Distinguished MS Dissertation Award at Seoul National University 2012
- Academic Performance Scholarship 2009
- National Science and Engineering Undergraduate Scholarship 2003–2008

SERVICE

- **Area Chair**
NeurIPS'24-26, ICML'25-26, ICLR'26-27
- **Reviewer**
NeurIPS'21-23, ICML'21-23, ICLR'22-24, AAAI'25, Journal of the Royal Statistical Society: Series B
- **External Reviewer**
S&P'21, S&P'22, Security'22, Security'23, Security'24, NDSS'24

TEACHING

- **Instructor** at POSTECH Fall 2025
ML (CSED515)
- **Instructor** at POSTECH Fall 2023, Fall 2024, Spring 2025
Trustworthy ML (AIGS703L / CSED703L)
- **Instructor** at POSTECH Spring 2024
Discrete Mathematics (CSED261)
- **Teaching Assistant** at University of Pennsylvania Spring 2015
Machine Perception (CIS580)
- **Teaching Assistant** at University of Pennsylvania Fall 2014
Computer Vision and Computational Photography (CIS581)
- **Teaching Assistant** at Seoul National University Fall 2010
Linear Algebra for Electrical Systems
- **Instructor** at Seoul National University Feb. 2008
1st Free Computer Education for Gwanak-gu Community Youth

TALKS

- **AI Alignment**
 Nxt AI Nov. 2025
 KAIST GSIS Nov. 2025
- **AI Red Teaming and Agentic AI Security**
 OSSTEM Nov. 2025
 Samsung SDS Nov. 2025
 WITHCON Nov. 2025
- **AI System for Discovering and Patching Vulnerable Code by Team Atlanta**
 POSCO DX Oct. 2025
- **Introduction to Trustworthy LLMs**
 KISC (한국통신학회 단기강좌) July. 2025
- **Trustworthy LLMs**
 Best of the Best (BoB) July. 2025
- **Selective Generation Towards Trustworthy LLMs**
 NYU Frontier Lab July. 2025
 KIISC AI Security Group (정보보호학회 AI보안 연구회) July. 2025
- **Toward Trustworthy Large Language Models**
 PKNU AI Dec. 2024
 POSTECH AI Day Dec. 2024
- **Rethinking and Harnessing Trustworthiness of Generative AI for Security**
 Samsung Security Tech Forum (SSTF) – *Invited Talk* Sep. 2024
- **Trustworthy Military-AI: AI Controllability**
 REAIM Sep. 2024
- **Trend on Trustworthy Language Models**
 National Statistics Development Forum Sep. 2024
- **Trustworthy AI: A Compositional Perspective**
 POSTECH AI/CSE April 2024
- **Conformal Prediction for Trustworthy AI**
 Korean AI Association Winter School Feb. 2024
- **Uncertainty Learning for Trustworthy and Secure AI**
 POSTECH AI/CSE Mar. 2023
 KAIST EE April 2023
 SNU CSE April 2023
 Korea University CSE July 2023
 SNU IPAI July 2023
 SNU Frontier Summer School Aug. 2023
 UNIST IB Oct. 2023
 KAIST EE Jan. 2024
 CAU AI May 2024
 Korea GSS June 2024
- **PAC Prediction Sets for AI Safety**
 ICML Workshop DFUQ 2022 – *Invited Talk* Jul. 2022
- **Uncertainty Quantification via PAC Prediction Sets**
 DGIST Dec. 2021
- **From Verification to Learning for Defense against Adversarial Examples in Neural Networks**
 KAIST GSIS Aug. 2018
 Hanyang University Aug. 2018
 KIISC Aug. 2018